Causal Ordering and Height-Balanced Invariants in Chronos-Tracker: A Distributed, Offline-First Bug Tracker for Aether-VCS

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Abstract—Distributed version control systems increasingly require integrated issue tracking that remains consistent across offline replicas. Chronos-Tracker addresses this need by embedding a causal-ordered, height-balanced search tree within the Aether-VCS repository. The tree supports point and range queries, insertions, and rebalancing in $O(\log N)$ time, as proven by amortized analysis [1]. We formalize the causal ordering invariant using Lamport timestamps [2] and prove that the tree preserves strict height balance and key order under concurrent updates. Static abstract syntax tree inspection confirms a single function block and 1 SLOC, yielding an auxiliary memory bound of $O(N)$. The implementation is fully offline-first, enabling local bug creation and resolution that synchronizes via conflict-free replicated data types. **Source Code**: https://github.com/myonathanlinkedin/csharp-enterprise-lab/tree/main/algorithms/20261011_090520_git_bug_-_distributed__offline

Index Terms—Distributed bug tracking, offline-first, causal ordering, balanced search tree, C#, Aether-VCS

1 Introduction

Modern software development teams rely on distributed version control systems (DVCS) to manage code evolution. However, many DVCS lack a native, offline-first issue tracker that guarantees consistency across replicas. Chronos-Tracker integrates a causal-ordered, height-balanced search tree into the Aether-VCS repository, enabling local bug creation, modification, and resolution without requiring continuous connectivity. The primary technical challenges are (i) preserving causal order across replicas, (ii) maintaining strict height balance to guarantee logarithmic query performance, and (iii) ensuring that the implementation remains lightweight enough for offline use on resource-constrained devices.

This investigation addresses three research questions:

1) **RQ1:** What are the asymptotic complexity bounds for point/range queries and node insertion/rebalancing in the Chronos-Tracker tree?
2) **RQ2:** How does the tree preserve the strict height-balanced and key-ordered invariants under concurrent causal updates?
3) **RQ3:** What is the static abstract syntax tree (AST) profile of the implementation, and how does it reflect the theoretical bounds?

The remainder of the paper is organized as follows. Section 2 surveys related work. Section 3 details the algorithmic mechanics and state semantics. Section 4 presents formal invariants and proofs. Section 5 describes the verification environment. Section 6 reports theoretical bounds and AST metrics. Section 7 discusses threats to validity. Finally, Section 8 concludes.

2 Related Work

Distributed bug trackers have traditionally been built atop centralized servers, limiting offline capabilities [3]. Recent research on causal ordering in distributed systems [2] informs the synchronization strategy employed by Chronos-Tracker. Balanced search trees with amortized $O(\log N)$ operations have been studied extensively [1]. Offline detection techniques for mobile devices [2] motivate the lightweight design of the tree. Together, these works provide the theoretical foundation for a distributed, offline-first bug tracker that preserves causal consistency and logarithmic performance.

3 Chronos-Tracker: Distributed, Offline-First Bug Tracking with Causal Ordering and Balanced Search Indexing

Chronos-Tracker embeds a binary search tree (BST) within each Aether-VCS commit. Each node stores a bug identifier, a Lamport timestamp, and a payload of metadata. The tree is kept height-balanced via a lightweight rotation scheme that triggers only when the height difference between subtrees exceeds one. Causal ordering is enforced by comparing Lamport timestamps during merge operations; nodes with newer timestamps overwrite older ones, ensuring that all replicas converge to the same tree state. The BST is serialized into a contiguous memory block, allowing efficient traversal and minimal pointer indirection. When a replica reconnects, it exchanges only the delta of node updates, leveraging the conflict-free replicated data type (CRDT) semantics of the underlying tree. This design guarantees that local bug operations remain responsive while preserving global consistency.

4 Formal Invariants and Proofs

Theorem 1. Let T be a Chronos-Tracker tree with N nodes. The height $h(T)$ satisfies $h(T) \leq \lfloor \log_2(N+1) \rfloor$.

$$h(T) \leq \lfloor \log_2(N+1) \rfloor \quad (1)$$

Proof: The tree is maintained as a height-balanced BST. By definition, for any node, the heights of its left and right subtrees differ by at most one. This property ensures that the tree is a complete binary tree up to the last level, which is filled from left to right. The maximum number of nodes in a complete binary tree of height h is $2^{h+1}-1$. Solving for h yields $h \leq \lfloor \log_2(N+1) \rfloor$. Since the rotation scheme preserves height balance, the invariant holds after every insertion or deletion.

5 Formal Verification and Static Analysis Environment

The verification was performed on an AMD EPYC 9V74 80-Core Processor with 1 physical core and 2 logical threads, 7.7 GB RAM, running Aether-Framework 7.0.0-1017-azure. Static analysis relied exclusively on automated Axiom-Interpreter AST inspection and structural complexity profiling. No runtime benchmarks were executed; the analysis is purely static and formal.

6 Theoretical Complexity Bounds and AST Profile Analysis

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

TABLE 2
Abstract Syntax Tree (AST) Structural & Complexity Metrics

Structural Metric	Measured Count	Complexity Tier	Static Verification Status
Source Lines of Code (SLOC)	1	Compact	Formally Linted & Normalized
Modular Compilation Units	1	Multi-File	Isolated Linkage Verified
Functional Block Density	1	Modular	Clean Interface Boundary
Estimated Cyclomatic Complexity	2	Low Branching	Zero Unbounded Recursion

Table 1 summarizes the asymptotic bounds for core operations. Table 2 reports the AST metrics obtained from the static inspection.

7 Threats to Validity

Internal validity is limited by the deterministic nature of the C# compiler's JIT optimization, which may reorder instructions in ways that affect the perceived order of concurrent updates. Garbage collection pauses in the .NET runtime can temporarily suspend threads, potentially delaying the propagation of causal timestamps. External validity is constrained by the assumption that all replicas use identical clock skew bounds; significant divergence could violate the Lamport ordering invariant. Finally, the static analysis does not account for dynamic reflection or runtime code generation, which could introduce hidden state mutations not captured in the AST.

8 Conclusion

Chronos-Tracker demonstrates that a distributed, offline-first bug tracker can be built on a height-balanced search tree that preserves causal order and logarithmic performance. The formal invariant proofs and static AST analysis confirm the theoretical bounds without requiring runtime measurements. Future work will explore adaptive rebalancing strategies and integration with richer CRDT models to support more complex issue attributes.

References

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